Overview of vitamins and minerals

How are vitamins classified?

A) Based on their molecular weight

B) Based on their solubility in water or fat

C) Based on their source (plant vs. animal)

D) Based on their role in DNA synthesis

ANSWER: B

Which vitamin is also known as ascorbic acid?

A) Vitamin B12

B) Vitamin C

C) Vitamin K

D) Vitamin E

ANSWER: B

Which mineral is classified as a major mineral (macromineral)?

A) Iron (Fe)

B) Zinc (Zn)

C) Calcium (Ca)

D) Iodine (I)

ANSWER: C

What is the active form of Vitamin B1 (Thiamine)?

A) Flavin adenine dinucleotide (FAD)

B) Thiamine pyrophosphate (TPP)

C) Nicotinamide adenine dinucleotide (NAD)

D) Pyridoxal phosphate (PLP)

ANSWER: B

Which of the following is a rich source of Vitamin A?

A) Egg and milk

B) Vegetable oils

C) Dark green leafy vegetables

D) Yeast

ANSWER: A

Which of the following is considered the primary function of Vitamin K?

A) Antioxidant

B) Blood clotting

C) Coenzyme in redox reactions

D) Collagen synthesis

ANSWER: B

What is the active form of Vitamin B2 (Riboflavin)?

A) Coenzyme A (CoA-SH)

B) Thiamine pyrophosphate (TPP)

C) Flavin mononucleotide (FMN)

D) Tetrahydrofolic acid (THF)

ANSWER: A

Which vitamin acts as a carrier for one-carbon moieties like methyl and formyl groups?

A) Vitamin B6

B) Vitamin C

C) Vitamin B12

D) Vitamin B9 (Folic acid)

ANSWER: D

Which vitamin deficiency during pregnancy increases the risk of fetal neural tube defects?

A) Vitamin B12

B) Vitamin B9 (Folic acid)

C) Vitamin B6

D) Vitamin D

ANSWER: B

What is the active form of Vitamin B6 (Pyridoxine)?

A) Pyridoxal phosphate (PLP)

B) Methylcobalamin

C) Nicotinamide adenine dinucleotide (NAD)

D) Biotin

ANSWER: A

Which of the following is a source of Vitamin B12 (Cobalamin)?

A) Whole grains

B) Nuts

C) Liver and meat

D) Vegetable oils

ANSWER: C

Which vitamin is essential for the formation of acyl carrier protein (ACP) needed for fatty acid synthesis?

A) Vitamin B5 (Pantothenic acid)

B) Vitamin B7 (Biotin)

C) Vitamin B3 (Niacin)

D) Vitamin B2 (Riboflavin)

ANSWER: A

What is the deficiency manifestation of Vitamin D in children?

A) Osteomalacia

B) Rickets

C) Osteopenia

D) Megaloblastic anemia

ANSWER: B

A patient taking methotrexate for cancer should consult their oncologist before taking supplements of which vitamin?

A) Vitamin B5

B) Vitamin B6

C) Vitamin B7

D) Vitamin B9

ANSWER: D

A 30-year-old pregnant woman is advised to increase her intake of dark green leafy vegetables. This is primarily to prevent neural tube defects in the fetus by ensuring adequate levels of:

A) Vitamin B12

B) Vitamin B9

C) Vitamin B6

D) Vitamin C

ANSWER: B

A patient with scurvy presents with spongy bleeding gums, loose teeth, and fragile blood vessels. Which of the following functions is impaired in this patient?

A) Biosynthesis of collagen

B) Blood clotting

C) Myelination

D) Antioxidant defense

ANSWER: A

A patient with a deficiency of Vitamin B1 affecting the cardiovascular system presents with edema. This condition is known as:

A) Dry beriberi

B) Wet beriberi

C) Pellagra

D) Scurvy

ANSWER: B

A patient with a deficiency of Cobalamin shows neurological manifestations. This is because Cobalamine is required for:

A) DNA synthesis

B) REDOX reactions

C) Collagen synthesis

D) Myelination and function of the CNS

ANSWER: D

A patient with a history of fat malabsorption presents with bleeding due to a deficiency of prothrombin. Which vitamin is most likely deficient?

A) Vitamin A

B) Vitamin D

C) Vitamin E

D) Vitamin K

ANSWER: D

A 70-year-old woman is diagnosed with osteomalacia. Which vitamin deficiency is the most likely cause?

A) Vitamin A

B) Vitamin D

C) Vitamin E

D) Vitamin K

ANSWER: B

A patient with a deficiency of Pyridoxine may present with convulsions and a pellagra-like syndrome. Which of the following is a function of Pyridoxine?

A) Essential for glycogen phosphorylation

B) Coenzyme in oxidative decarboxylation of pyruvate

C) Carrier of one-carbon moieties

D) Antioxidant

ANSWER: A